DEEP Detailed Assessment

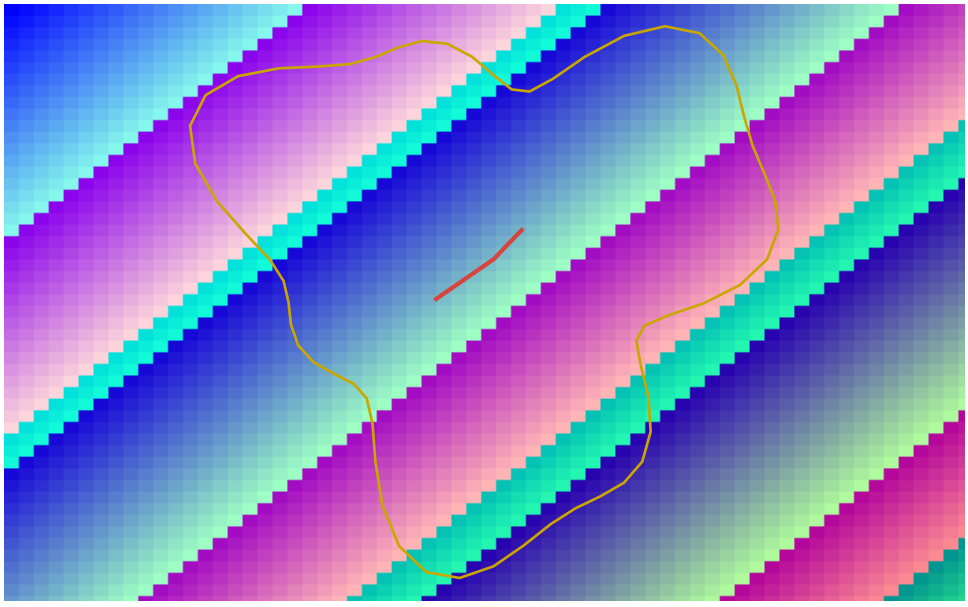
Northeastern Highlands reference assessment

(unnamed reach)

Scenario: Existing Conditions

Source	USEPA NRSA 2013-24 (L3 ecoregion 58), StreamCurves regional analysis, methodology 0.16
Version / status	v10 · Preliminary
Predictor source	streamcat
Region match	58 Northeastern Highlands
Coordinates	43.7, -72.2
COMID / HUC8	9327042 / 01080104
Drainage area	3.0 km2
HR reach watershed area	11.98 km2
Reach length	1234.5 ft
Watershed basis	NHDPlus V2 basin of the nearest StreamCat reach (StreamCat lookup engine)
Desktop values	STAF site engine (HR reach watershed) 5, StreamCat lookup engine (NHDPlus V2 basin) 2
StreamCat reach	COMID 9327042 (this reach)
Content digest	sha256:3803702732ec6e7191238f165ffae1d5e6588465a682a891fbf58f3d1e2256ca
Ecosystem Condition Index	0.41 (Functioning-at-Risk)
STAF function coverage	20 of 20
Flagged transfers	1 of the 20 rated functions rest on a reference pool whose transfer to this ecoregion was not confirmed by the recovery test (Surface water storage); each is rated with confidence capped and flagged on its metric row.

Watershed



Function scores (0-15)

Function	Score	Condition
Catchment hydrology	13.3	F
Surface water storage		Not scored
Reach inflow	5.9	AR
Streamflow regime	8.5	AR
Low flow and baseflow dynamics	8.4	AR
High flow dynamics	13.6	F
Floodplain connectivity	14.1	F
Hyporheic connectivity		Not scored
Channel evolution	14.1	F
Channel and floodplain dynamics	10.5	F
Sediment continuity		Not scored
Bed composition and bedform dynamics		Not scored
Light & thermal regime	5.4	AR
Carbon processing		Not scored
Nutrient cycling		Not scored
Water & soil quality	0.0	NF
Habitat provision		Not scored
Population support		Not scored
Community dynamics	0.0	NF
Watershed connectivity	0.0	NF

Function limitations

Surface water storage: This function is rated on a curve whose reference pool transfer to this ecoregion was not confirmed by the recovery test (Wetland); the score enters the index with confidence capped.

Outcome sub-indices

Physical	Chemical	Biological	ECI
0.71	0.33	0.21	0.41

Metric measurements & curve indices

Function	Metric	Value	Index	Source	Curve (source)
Catchment hydrology	Impervious surface	3.0	0.91	desktop: STAF site engine v0.5.0 impervious (HR reach watershed, NLCD 2021)	EASI fixed criteria, Catchment hydrology (land-cover pressure)
Catchment hydrology	Agriculture, crop and hay	12.75	0.87	desktop: StreamCat PCTIMP2019WS	EASI fixed criteria, Catchment hydrology (land-cover pressure)

Function	Metric	Value	Index	Source	Curve (source)
Surface water storage	Wetland	0.4	ref. only	desktop: Computed (reference only)	USEPA NRSA 2013-24 (L3 ecoregion 58), StreamCurves regional analysis, methodology 0.16-provisional
Reach inflow	Road density	3.0	0.39	desktop: STAF site engine v0.5.0 impervious (HR reach watershed, NLCD 2021)	EASI fixed criteria, Road-density proxy for concentrated inflow pressure
Streamflow regime	Base flow index	12.75	0.19	desktop: StreamCat PCTIMP2019WS	USEPA NRSA 2013-24 (L3 ecoregion 58), StreamCurves regional analysis, methodology 0.16-provisional
Streamflow regime	Degree of regulation	0.4	0.94	desktop: Computed	EASI fixed criteria, Flow alteration (degree of regulation)
Low flow and baseflow dynamics	Base flow index	12.75	0.19	desktop: StreamCat PCTIMP2019WS	USEPA NRSA 2013-24 (L3 ecoregion 58), StreamCurves regional analysis, methodology 0.16-provisional
Low flow and baseflow dynamics	Bankfull width:depth ratio	7.5	0.94	field	USEPA NRSA 2013-24 (L3 ecoregion 58), StreamCurves regional analysis, methodology 0.16-provisional
High flow dynamics	Impervious surface	3.0	0.91	desktop: STAF site engine v0.5.0 impervious (HR reach watershed, NLCD 2021)	EASI fixed criteria, Catchment hydrology (land-cover pressure)
Floodplain connectivity	Bankfull width:depth ratio	7.5	0.94	field	USEPA NRSA 2013-24 (L3 ecoregion 58), StreamCurves regional analysis, methodology 0.16-provisional
Floodplain connectivity	Bankfull height above channel				USEPA NRSA 2013-24 (L3 ecoregion 58), StreamCurves regional analysis, methodology 0.16-provisional
Hyporheic connectivity	Sand + fines				USEPA NRSA 2013-24 (L3 ecoregion 58), StreamCurves regional analysis, methodology 0.16-provisional
Hyporheic connectivity	Embeddedness			field	USEPA NRSA 2013-24 (L3 ecoregion 58), StreamCurves regional analysis, methodology 0.16-provisional
Channel evolution	Bankfull width:depth ratio	7.5	0.94	field	USEPA NRSA 2013-24 (L3 ecoregion 58), StreamCurves regional analysis, methodology 0.16-provisional
Channel evolution	Bankfull height above channel				USEPA NRSA 2013-24 (L3 ecoregion 58), StreamCurves regional analysis, methodology 0.16-provisional
Channel and floodplain dynamics	Sinuosity				USEPA NRSA 2013-24 (L3 ecoregion 58), StreamCurves regional analysis, methodology 0.16-provisional

Function	Metric	Value	Index	Source	Curve (source)
Channel and floodplain dynamics	Bank angle	12.5	0.70	field	USEPA NRSA 2013-24 (L3 ecoregion 58), StreamCurves regional analysis, methodology 0.16-provisional
Sediment continuity	Relative bed stability				USEPA NRSA 2013-24 (L3 ecoregion 58), StreamCurves regional analysis, methodology 0.16-provisional
Sediment continuity	Sand + fines				USEPA NRSA 2013-24 (L3 ecoregion 58), StreamCurves regional analysis, methodology 0.16-provisional
Bed composition and bedform dynamics	Substrate size				USEPA NRSA 2013-24 (L3 ecoregion 58), StreamCurves regional analysis, methodology 0.16-provisional
Bed composition and bedform dynamics	Large wood volume				USEPA NRSA 2013-24 (L3 ecoregion 58), StreamCurves regional analysis, methodology 0.16-provisional
Light & thermal regime	Chlorophyll a				USEPA NRSA 2013-24 (L3 ecoregion 58), StreamCurves regional analysis, methodology 0.16-provisional
Light & thermal regime	Canopy density, mid-channel	17.5	0.36	field	USEPA NRSA 2013-24 (L3 ecoregion 58), StreamCurves regional analysis, methodology 0.16-provisional
Carbon processing	Riparian veg cover, woody				USEPA NRSA 2013-24 (L3 ecoregion 58), StreamCurves regional analysis, methodology 0.16-provisional
Nutrient cycling	Total nitrogen				USEPA NRSA 2013-24 (L3 ecoregion 58), StreamCurves regional analysis, methodology 0.16-provisional
Nutrient cycling	Total phosphorus				USEPA NRSA 2013-24 (L3 ecoregion 58), StreamCurves regional analysis, methodology 0.16-provisional
Water & soil quality	Specific conductivity				USEPA NRSA 2013-24 (L3 ecoregion 58), StreamCurves regional analysis, methodology 0.16-provisional
Water & soil quality	Turbidity	22.5	0.00	field	USEPA NRSA 2013-24 (L3 ecoregion 58), StreamCurves regional analysis, methodology 0.16-provisional
Habitat provision	Fast-water / riffle habitat				USEPA NRSA 2013-24 (L3 ecoregion 58), StreamCurves regional analysis, methodology 0.16-provisional
Habitat provision	Natural fish cover			field	USEPA NRSA 2013-24 (L3 ecoregion 58), StreamCurves regional analysis, methodology 0.16-provisional

Function	Metric	Value	Index	Source	Curve (source)
Population support	Benthic total taxa richness				USEPA NRSA 2013-24 (L3 ecoregion 58), StreamCurves regional analysis, methodology 0.16-provisional
Community dynamics	Shannon diversity (H')				USEPA NRSA 2013-24 (L3 ecoregion 58), StreamCurves regional analysis, methodology 0.16-provisional
Community dynamics	Tolerant individuals (%) ~ HBI	27.5	0.00	field	USEPA NRSA 2013-24 (L3 ecoregion 58), StreamCurves regional analysis, methodology 0.16-provisional
Watershed connectivity	Mapped dams within one mile	3.0	0.00	desktop: STAF site engine v0.5.0 impervious (HR reach watershed, NLCD 2021)	EASI fixed criteria, Nearby dam-proximity proxy

Scoring advisories

Wetland: engine-computed value shown as reference only: this curve was fitted on StreamCat predictors, so scoring it against an HR reach watershed value would mix training and serving sources

Base flow index: value 12.75 is below the reference pool's range (42.982 to 62); the score follows the seed's sub-reference convention, not reference data

Base flow index: value 12.75 is below the reference pool's range (42.982 to 62); the score follows the seed's sub-reference convention, not reference data

Turbidity: value 22.5 is above the curve domain [0.0, 3.975]; score clamped to the endpoint; value 22.5 is above the reference pool's range (0.01 to 7.52); the score follows the seed's convention beyond the data

Tolerant individuals (%) ~ HBI: value 27.5 is above the curve domain [0.0, 24.315]; score clamped to the endpoint

Reference support

Reference condition is set by a fixed landscape-pressure screen, never by a score. 28 curves are fitted to least-disturbed stations of this ecoregion. 1 borrows comparable least-disturbed stations from a wider region, stating the region and the transfer risk. 2 carry the national reference label: reference curve fitted to comparable least-disturbed stations drawn from the national pool, because this ecoregion and its parents hold too few of their own. 2 rest on a flagged transfer: a comparable least-disturbed pool the recovery test did not confirm for this ecoregion, used because no validated source supports the metric here, with transfer risk unvalidated and confidence capped. A metric that no basis supports is not scored, and the function it would have scored is reported as unassessed.

Metric	Scored against	Curve set
Impervious surface	Fixed criteria, the same in every region	
Agriculture, crop and hay	Fixed criteria, the same in every region	
Wetland	Reference curve fitted to comparable least-disturbed stations drawn from the national pool, because this ecoregion and its parents hold too few of their own.	
Road density	Fixed criteria, the same in every region	
Base flow index	Reference curve from 66 least-disturbed stations of this ecoregion	
Degree of regulation	Fixed criteria, the same in every region	
Bankfull width:depth ratio	Reference curve from 66 least-disturbed stations of this ecoregion	All streams (pooled) (default)
Bankfull height above channel	Reference curve from 66 least-disturbed stations of this ecoregion	All streams (pooled) (default)
Sand + fines	Reference curve from 66 least-disturbed stations of this ecoregion	
Embeddedness	Reference curve from 63 least-disturbed stations of this ecoregion	
Sinuosity	Reference curve from 63 least-disturbed stations of this ecoregion	
Bank angle	Reference curve from 98 least-disturbed stations borrowed from Level II ecoregion 5.3 (Atlantic Highlands) under the regional screen (watershed agriculture at most 25 percent), 90 of them in this ecoregion. Transfer risk low	
Relative bed stability	Reference curve from 65 least-disturbed stations of this ecoregion	All streams (pooled) (default)
Substrate size	Reference curve from 66 least-disturbed stations of this ecoregion	
Large wood volume	Reference curve from 50 least-disturbed stations of this ecoregion	
Chlorophyll a	Reference curve from 66 least-disturbed stations of this ecoregion	
Canopy density, mid-channel	Reference curve from 63 least-disturbed stations of this ecoregion	All streams (pooled) (default)

Metric	Scored against	Curve set
Riparian veg cover, woody	Reference curve from 66 least-disturbed stations of this ecoregion	
Total nitrogen	Reference curve from 66 least-disturbed stations of this ecoregion	
Total phosphorus	Reference curve from 65 least-disturbed stations of this ecoregion	
Specific conductivity	Reference curve from 66 least-disturbed stations of this ecoregion	
Turbidity	Reference curve from 62 least-disturbed stations of this ecoregion	
Fast-water / riffle habitat	Reference curve from 66 least-disturbed stations of this ecoregion	All streams (pooled) (default)
Natural fish cover	Reference curve from 66 least-disturbed stations of this ecoregion	All streams (pooled) (default)
Benthic total taxa richness	Reference curve from 65 least-disturbed stations of this ecoregion	
Shannon diversity (H')	Reference curve from 65 least-disturbed stations of this ecoregion	
Tolerant individuals (%) ~ HBI	Reference curve from 65 least-disturbed stations of this ecoregion	
Mapped dams within one mile	Fixed criteria, the same in every region	

Metrics not scored

These metrics are not scored. A metric withheld for insufficient reference support has no curve. No pool of least-disturbed stations in this ecoregion or its parents supports it, no national donor pool was shown to transfer, no modeled expectation passed validation, and no published criterion applies.

Metric	Functions	Reason
Native fish taxa richness	Population support	Insufficient reference support
Residual pool depth	Low flow and baseflow dynamics	Insufficient reference support

Scores are computed automatically from the assessment's reference curves. Confirm state/region applicability of the curve source.